
Radar Altimeter

US-D1 Pro

User Manual

D00.00.07
August 22, 2025

Revision History

Version Number	Date	Notes
D00.00.01	Mar 28, 2025	Initial
D00.00.02	April 14, 2025	Revise specs for A sample
D00.00.03	July 1, 2025	Add Section of Hardware Interface, Communication Protocol, Mechanical Specification.
D00.00.04	August 4, 2025	Add detailed information in Private CAN and DroneCAN in Section 6
D00.00.05	August 5, 2025	Update Specs for US-D1 Pro B sample
D00.00.06	August 18, 2025	Update Specs for US-D1 Pro B sample firmware v1.0
D00.00.07	August 22, 2025	Add the section of Firmware Update

1. Product Description

1.1. US-D1 Pro

The US-D1 Pro is the second generation of **short-mid range Radar Altimeter**. It uses the principles of radio detection and ranging to determine the altitude of the aircraft. A microwave signal is transmitted out of the sensor, reflects off the target, and is received by the sensor. Distance from the sensor to the terrain (altitude) is derived from the difference in time from when the signal is sent from the sensor to when the signal is received by the sensor.

1.2. Compliances

1.2.1 Radio Equipment, EMC/EMI, RF

This equipment is designed to comply with the limits for a Class B device (commercial or industrial use), pursuant to Part 15 of the FCC rules or RED of 2014/53/EU. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communication. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at his/her own expense.

However, we have not performed the official test and have not obtained the grant of FCC ID from the authorities or don't have the CE Declaration of conformity ready yet.

1.2.2 Radio Technical Commission for Aeronautics (RTCA)

This equipment is designed to comply with RTCA DO-160, 178 and 254.

1. Environmental (DO-160) - This equipment is designed to comply with the RTCA DO-160G. However, we have not performed the official test and have not obtained the certification yet.

2. Software (DO-178) - This equipment is designed to comply with the RTCA DO-178C DAL C. However, we have not performed the official test and have not obtained the certification yet.

Electronic Hardware (DO-254) - This equipment is designed to comply with the RTCA DO-254 DAL C. However, we have not performed the official test and have not obtained the certification yet.

2. Installation Guidelines

Warning! Follow the manufacturer's regulations when mounting, modifying, repairing, and maintaining equipment. The manufacturer assumes no responsibility for any accidents caused by incorrectly mounted or incorrectly maintained equipment. The manufacturer assumes no responsibility for the system being incorrectly applied, or the system being programmed in a manner that jeopardizes safety.

The following instructions are critical to the proper operation of the US-D1 Pro device. Failure to install the unit according to the given instructions can result in malfunction of the device.

2.1. Mounting Angle and Position

When mounting the device, the front face of the US-D1 Pro beam must be **perpendicular** to the ground below the aircraft. There should not be any angle of inclination in any direction while the device is fastened to the aircraft.

In addition to the mounting angle, it is recommended that the angle between radar beam and other obstacle (e.g. landing gear) is greater than 50° and material is low permittivity type, to achieve the optimal performance.

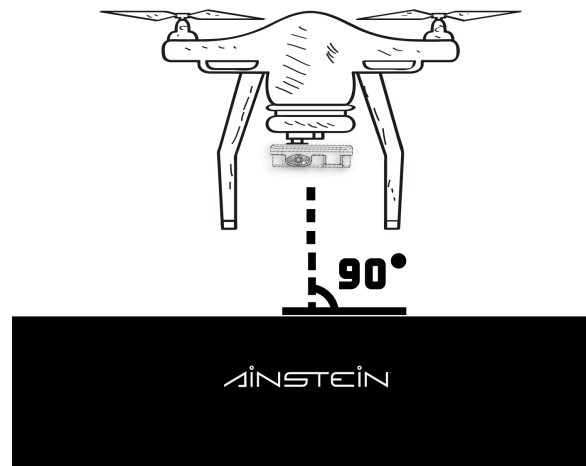


Figure 2.1

2.2 Mounting to an Aircraft

The device should be secured to the aircraft, where it is not free to move in any direction.

2.3. Line of Sight Clearance

Keep the face of the radar clean, and do not cover it with any additional materials. Any coatings, coverings, and modifications to the radome can degrade the performance of the radar device.

Additionally, keep any unexpected objects out of the radar's FoV (Field of View). Obstructions to the US-D1 Pro's field of view will cause a decrease in the performance of the radar. It is highly recommended that the angle between radar beam and expected objects (e.g. landing gear) greater than 50° and material is **low permittivity type**, to achieve the optimal performance.

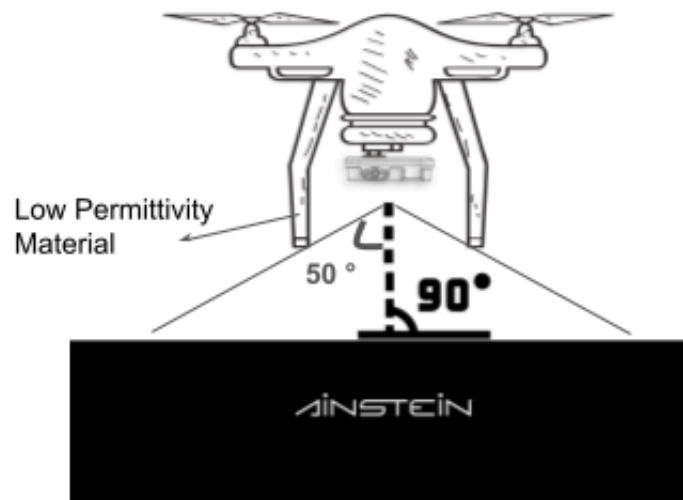


Figure 2.2

2.4. Integration Requirements

The US-D1 Pro outputs altitude (AGL) measurements, vertical speed measurement, signal-to-noise ratio (SNR), Measurement Confidence when operational. When integrating the US-D1 Pro radar altimeter, it is recommended to use Measurement Confidence to properly filter out erroneous altitude values.

A filtering algorithm should be used to estimate vehicle position, velocity and angular orientation based on rate gyroscopes, accelerometer, compass, GPS, airspeed and barometric pressure measurements in addition to the recorded US-D1 Pro measurements. Sensor redundancy is heavily advised for the US-D1 Pro.

3. Operational Requirements

The US-D1 Pro will perform optimally if the operational requirements below are fully satisfied. Failure to meet the operational requirements may cause a decrease in performance, accuracy, or reliability of the US-D1 Pro altimeter and is not advised.

3.1 Obstruction to the US-D1 Pro Field of View

Objects or aircraft structures that are located within the US-D1 Pro's conical field of view which obstruct the radar's view of the ground may cause multipath reflections or other degradative phenomena to occur. The US-D1 Pro should be mounted a safe distance away from the landing gear and any other components of the aircraft below it.

3.2 Excessive Pitch/Roll

Pitch and roll angles that exceed 20° and 20° respectively may also cause the performance of the US-D1 Pro to worsen. This sensitivity increases with altitude.

3.3 Terrain

Terrain with poor reflectivity may cause the performance of the US-D1 Pro to worsen. Flying at the limit of the US-D1 Pro's range over dry, loose soil, such as tilled farmland, or sand is not recommended.

3.4 Power Source

Any power source used to operate the US-D1 Pro that does not provide the proper power and voltage can worsen the performance of the device or cause its operation to stop altogether. Moreover if the power voltage is significantly more than the input voltage upper limit, it may permanently damage the device.

3.5 Orientation of US-D1 Pro

The US-D1 Pro must be mounted upon the aircraft in such a manner as to be fully horizontal, its radome directly facing the ground. Any angle of inclination may degrade performance.

3.6 Minimum and Maximum Operating Altitude

Operating the US-D1 Pro at altitudes below its minimum altitude and above 120 meters will result in a degradation of performance and potentially erroneous measurements.

4. Technical Specifications

Table 1: Specifications

Operating Altitude (Minimum ~ Maximum)	The Minimum Range~120m ¹
Altitude Accuracy	±0.1m(@≤10m); 1% (@>10m)
Operating Velocity (Vertical)	-6.9 ~ +6.9m/s ²
Field of view (FOV)	Azimuth ±20° Elevation ±20°
Frequency Band	24 GHz ISM Band
Transmit ERIP	13 dBm ³
Update Rate	20Hz
Recommended Supply Voltage	DC 12~14V
Supply Voltage Range	DC 9~16V
Harness/Connector	MOLEX-349671001
Power consumption	3.5W ⁴
Temperature Range	-40°C~85°C (Operational) -55°C~85°C (Storage)
Weight	140 g
Size	101 mm * 76mm * 24.71 mm
Reliability	Total lifetime operating time: 6,000 hours, reliability: 0.98, operating failure rate: 3.5E-06 per hour. Total lifetime storage time: 60,000 hours, reliability: 0.98, storage failure rate: 3.8E-07 per hour. Total lifetime start-up cycles: 4,000 cycles, reliability: 0.98.

Interface	Two CAN-FD channels 1st channel for Private CAN message and OTA upgrade ⁵ 2nd channel for DroneCAN message and features ⁶
OTA Upgrade Method	- Private CAN by Ainstein's GUI - DroneCAN by tools supports it (DroneCAN GUI, Mission Planner, etc)
Protection level	IP69K
Time Synchronization	- DroneCAN - Supported in the node status reporting - Private CAN - NOT supported

Notes:

1. This is the Recommended AGL (Above Ground Level) altitude range. The minimum range depends on the actual mounting height and radar's minimum detection. For now, this minimum range is set as 0.65 meter.
2. This is the maximum unambiguous velocity range. It is recommended to use the velocity detection when the vehicle's vertical velocity within this range.
3. EIRP is the total transmitter power from US-D1 Pro, including the Transmitting Antenna Gain
4. This is the mean power consumption in steady operation. The peak power consumption might be greater than it.
5. OTA upgrade via Private CAN is performed by Ainstein GUI via wired method
6. Supported DroneCAN feature and functionality includes, but is not limited to (See details in Section 6.2):
 - a. Export Node Status message
 - b. Export Sensor Measurement message
 - c. Time synchronization
 - d. Node parameter configuration
 - e. Firmware update

5. Communication Protocol For US-D1 Pro

US-D1 Pro has two channels of CAN interface in hardware and it does support two CAN communication protocols: Private CAN and DroneCAN.

5.1 Private CAN

5.1.1 Private CAN Specification

- CAN Interface Type - Default of CAN 2.0
- Baud Rate - Default of 1 Mb/s in CAN 2.0
- Support Frame Format and ID - Default of Extended Frame with 0x90002
- Standard: CAN FD (maintains backward compatibility with CAN 2.0), ISO 11898-1

The type of CAN frame used and CAN ID of the device can be customized. The CAN frame Data Field is defined in the table below.

From	US-D1 Altimeter	To	Receiver
Byte	Data	Note	
1	0x**	Altitude (Most Significant 8 Bits) ¹	
2	0x**	Altitude (Least Significant 8 Bits)	
3	0x**	SNR(Most Significant 8 Bits) ²	
4	0x**	SNR(Least Significant 8 Bits)	
5	0x**	Confidence ³	
6	0x**	Velocity ⁴	
7	0x** (4-bit)	Counter ⁵	
8	(Reserved)	(Reserved)	

Table 5.1: Private CAN Data Format Definition

Notes:

1. Altitude: It is the AGL altitude reading. Unit: centimeter (0.01 m); Unsigned value;
2. SNR: It is the Signal-to-Noise Ratio for the reflection signal. Unit: dB; Unsigned value; Note: there is no Absolute Threshold in SNR for users' reference
3. Confidence: It indicates the radar's measurement confidence in percentage, its range is from 0 to 100. Unit: %.
 - a. Absolute threshold: 0 - Invalid detection

- b. Other: 1 ~ 100 - Reference for users
4. Velocity: It is the vertical velocity (ascending and descending); Unit: 0.1 m/s (0.1 meter per second); Range from -6.9 m/s to +6.9 m/s;
 5. Counter: It indicates the CAN message's counter, and automatically increments by 1 for a new CAN message. Range from 0 to 15;

5.1.2 Private CAN Configuration Options

No.	Parameter	Configurable Range	Default
1	CAN Interface Type	• CAN 2.0 • CAN FD	CAN 2.0
2	Frame Format	• Standard Frame • Extended Frame	Extended Frame
3	ID	• Standard Frame: 0x000 ~ 0x7FF; • Extended Frame: 0x00000000 ~ 0x1FFFFFFF	0x90002
4	Baud Rate	• CAN 2.0: ○ 500 Kbps ○ 1000 Kbps • CAN FD: ○ 500 Kbps ○ 1000 Kbps	1000 Kbps
5	Bit Rate Switch	• CAN FD: ○ 500 Kbps ○ 1000 Kbps ○ 2000 Kbps ○ 5000 Kbps	1000 Kbps

Table 5.2: Private CAN Configurable Parameter

5.2 DroneCAN Specification

- CAN Interface Type: CAN 2.0
- Baud Rate – Default of 1 Mb/s
- CAN Standard: CAN 2.0B
- Transfer Priority – Default of 24

- Source Node ID – Default ID 32
- Node Status Message
 - Data type ID: 341
 - Broadcasting in 1 Hz
- Range Sensor Measurement
 - Data type ID: 1050
 - Broadcasting in 20 Hz
 - Sensor ID
 - Default of 1
 - Configuration Range: 0-255

Compatible to Ardupilot and PX4 DroneCAN driver. Follow the official instruction to make the autopilot configuration.

5.2.1 DroneCAN Node Status

US-D1 Pro DroneCAN node status spec	DroneCAN message field	Setting	
Time Synchronization	uptime_sec	1 second	
Node Health	health		
		HEALTH_OK	Radar in normal operation
		HEALTH_WARNING	Radar in normal operation. Users shall use it in caution
		HEALTH_ERROR	Radar in stress operation. Users shall stop using it.
		HEALTH_CRITICAL	Radar failed to report. Users shall stop using it.

Current Node Status	mode		
		MODE_OPERATIONAL	Radar runs in Application firmware and in operation. It reports detection data and messages.
		MODE_INITIALIZATION	Radar is booting, no data reporting.
		MODE_MAINTENANCE	Radar runs in Bootloader firmware and no detection data and messages.
		MODE_SOFTWARE_UPDATE	Radar is under Firmware update
		MODE_OFFLINE	N.A.
Sub mode - Not used currently	sub_mode	0	
Optional, vendor-specific node status code	vendor_specific_status_code	0	

Table 5.3: Public CAN (DroneCAN) Node Status Data Format Definition

5.2.2 DroneCAN Sensor Measurement

US-D1 Pro DroneCAN sensor measurement spec	DroneCAN message field	Setting
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Timestamp for synchronization	timestamp	US-D1 Pro is a Time Slave node, it synchronizes the timestamp with the Time Master node on the bus. If there is no master clock node, it reports 0
Sensor ID	sensor_id	Default in 1; Configurable
Beam Orientation	beam_orientation_in_body_frame	N.A. Fixed in 0
Field of View	field_of_view	Note: - US-D1 Pro 'SNR' data in dB filled in this field - It may be changed to 'Confidence' in the future firmware as it is a more reliable parameter for the user to determine the validity of the radar's detection
Radar's Sensor Type	sensor_type	3
Radar's Reading Type	reading_type	It is fixed as 1 in all cases
Range	Range	US-D1 Pro range data in meters

Table 5.4: Public CAN (DroneCAN) Sensor Measurement Data Format Definition

5.2.3 DroneCAN Configuration Option

No.	Parameter	Configurable Range	Default
1	Transfer Priority	0~31	24
2	Node ID	0~127 0 - Reserved for DNA 126, 127 - Reserved for debug	32

3	Sensor ID	0~255	1
4	Baud Rate	• CAN 2.0: ○ 500 Kbps ○ 1000 Kbps • CAN FD: ○ 500 Kbps ○ 1000 Kbps	1000 Kbps

Table 5.5: DroneCAN Configurable Parameter

NOTE:

- When connecting the US-D1 Pro to ArduPilot, users must ensure that the RNGFNDX_ADDR parameter matches the sensor ID of the radar (i.e., if Sensor ID = 1, then RNGFNDX_ADDR = 1)

6. Hardware Interface

US-D1 Pro's connector and its receptacle are automotive grade and have the IP69K rating.

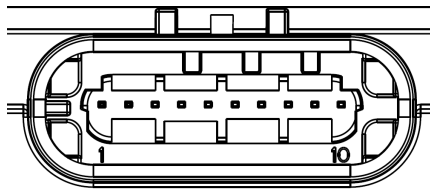


Figure 5.1: US-D1 Pro connector

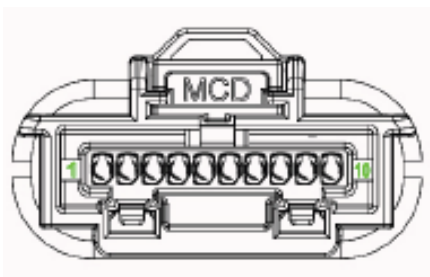


Figure 6.2: US-D1 Pro Connector's Receptacle - MOLEX-349671001

Pin	Definition	Function	Remark
1	CANL_PRI	Private CAN-L low	For Private CAN channel, an external 120Ω resistor is required
2	CANH_PRI	Private CAN-H	

		high	
3	CANL_PUB	Public CAN-L low	For the DroneCAN channel; 120Ω resistor requirement is up to the bus setting
4	CANH_PUB	Public CAN-H high	
5	NC	none	
6	NC	none	
7	NC	none	
8	GND_IN	Power ground	The power ground and signal ground are short-circuited and isolated from the shell (to improve ESD anti-interference ability)
9	NC	none	
10	DC_IN	Power input	9V~16V DC Input

Table 6.1: Pinout Definition

7. Mechanical Specifications

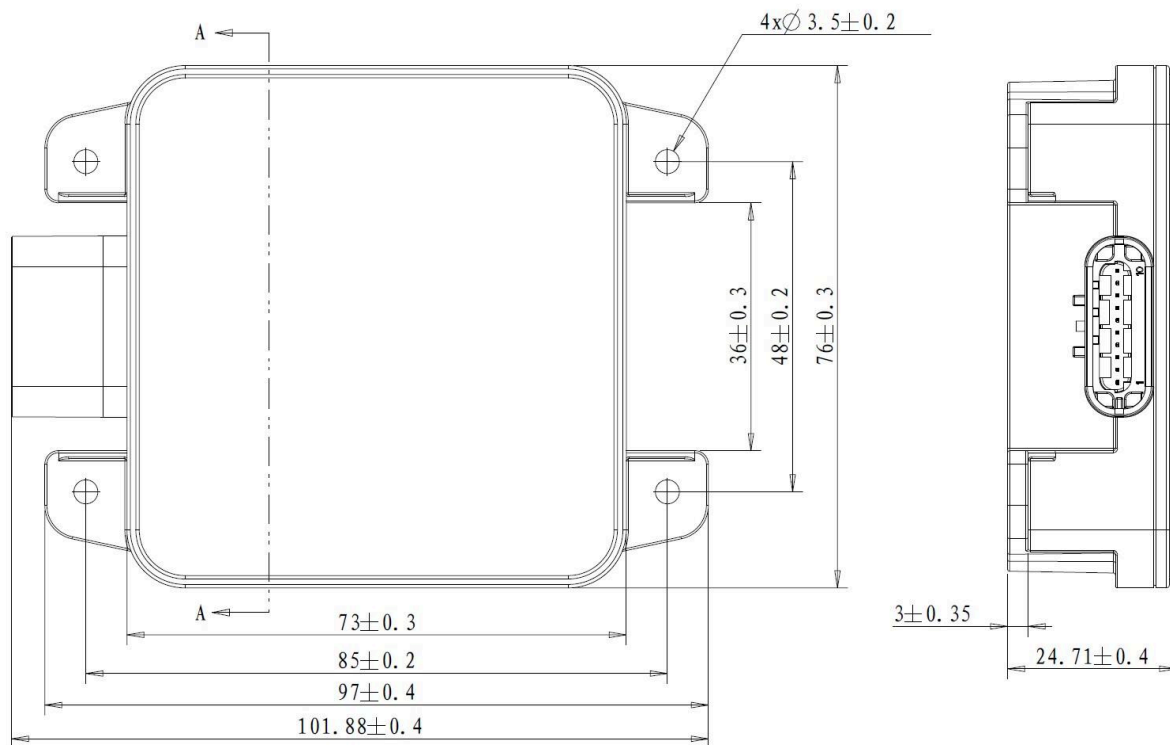


Figure 7.1: US-D1 Pro Dimension (Unit: mm)

AINSTEIN

1421 Research Park Dr

Lawrence, KS 66049

8. Firmware and Parameter Update

8.1 Private CAN Firmware and Parameter Update

To perform the firmware and parameter update via Private CAN, Ainstein will provide the following tool and file.

1. US-D1 Pro OTA Updater GUI Tool
2. Firmware Flash File for Firmware Update (.bin)
3. Data Identifier (DID) sheet for Parameter Configuration

Users should have the following tools.

1. CAN-FD Adapter
2. PC in Windows OS

Note:

- The CAN transceiver in US-D1 Pro hardware is CAN-FD, and it is compatible and can be configurable to CAN 2.0.
 - The interface between this US-D1 Pro OTA Updater GUI Tool and US-D1 Pro is fixed in CAN-FD, instead of CAN2.0. Therefore a CAN-FD Adapter is required.

Please see the configurable parameter in Section 5.1.2, and contact Ainstein for more assistance.

8.1.1 'US-D1Pro OTA Updater' and Firmware Update Instruction

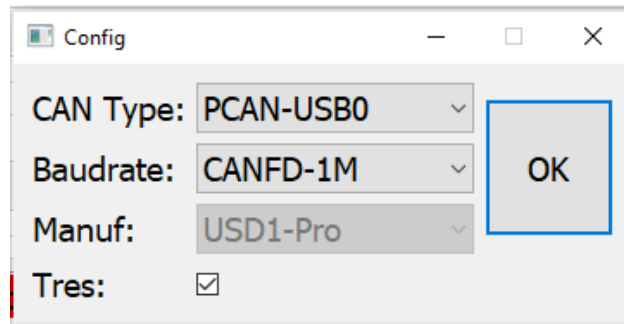
US-D1 Pro OTA Updater is the GUI tool to perform the firmware and parameter update via Private CAN, here are the steps:

1. Open the tool and enter the tool configuration window
2. Select the connection interface, and click 'OK' to enter the main window

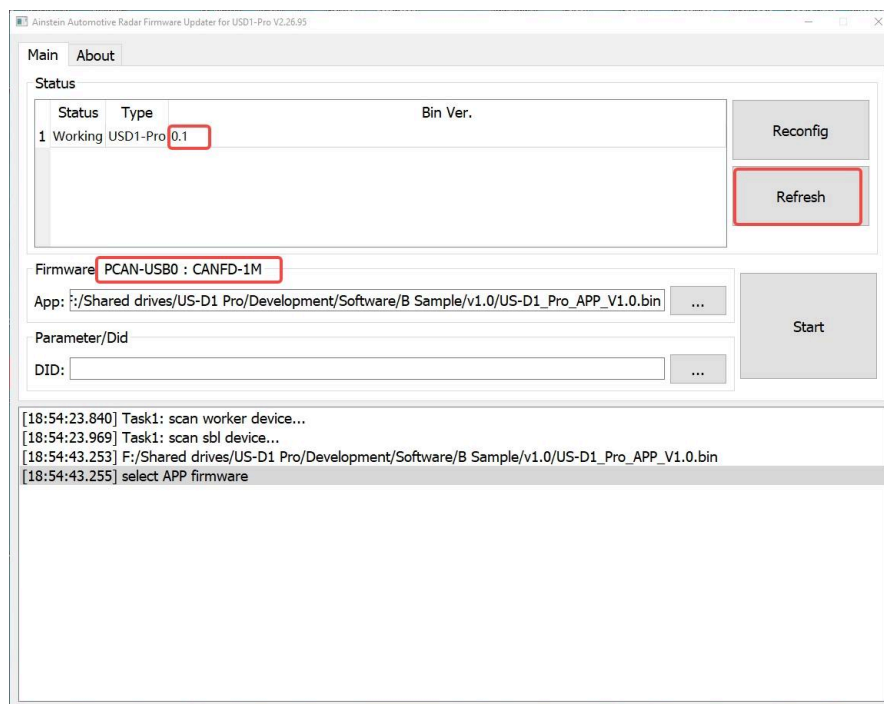
Note:

- The interface between this GUI and US-D1 Pro is fixed in CAN-FD, instead of CAN2.0, please select 'CANFD-XX' in Baudrate option. This doesn't have the impact on the US-D1 Pro streaming data's interface and baud rate. Example:
 - If US-D1 Pro has been configured to 500Kbps in Baud rate, select CANFD-500k; while in the detection data stream export, it can be in CAN-FD or CAN 2.0 at 500Kbps.

- If US-D1 Pro has been configured to 1Mbps in Baud rate, select CANFD-1M; while in the detection data stream export, it can be in CAN-FD or CAN 2.0 at 1Mbps.
- CAN type depends on the user's setting. In this example, CAN type is selected as 'PCAN-USB0'



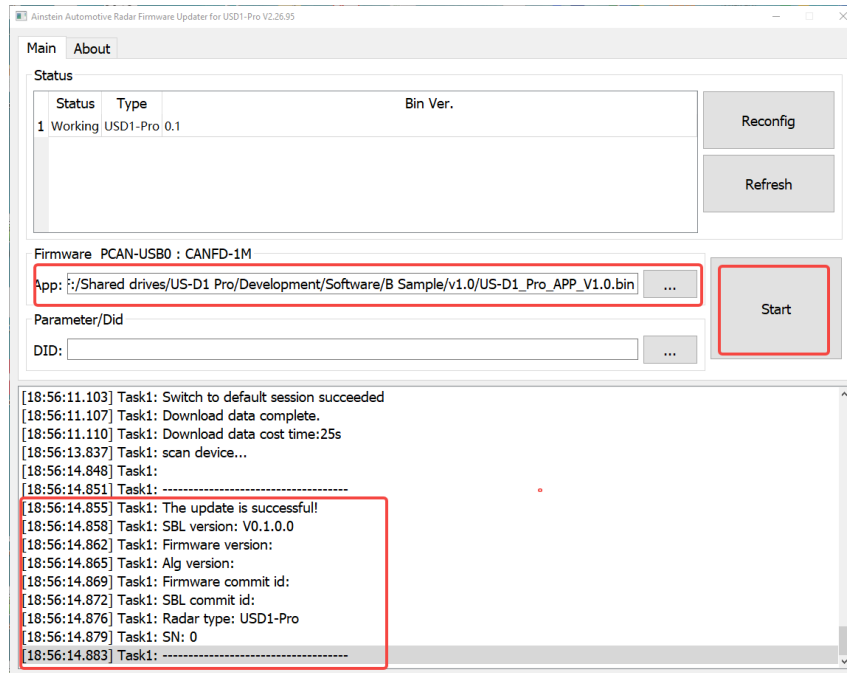
3. In the main window, click 'Refresh' to inquire about US-D1 Pro's current status and firmware version



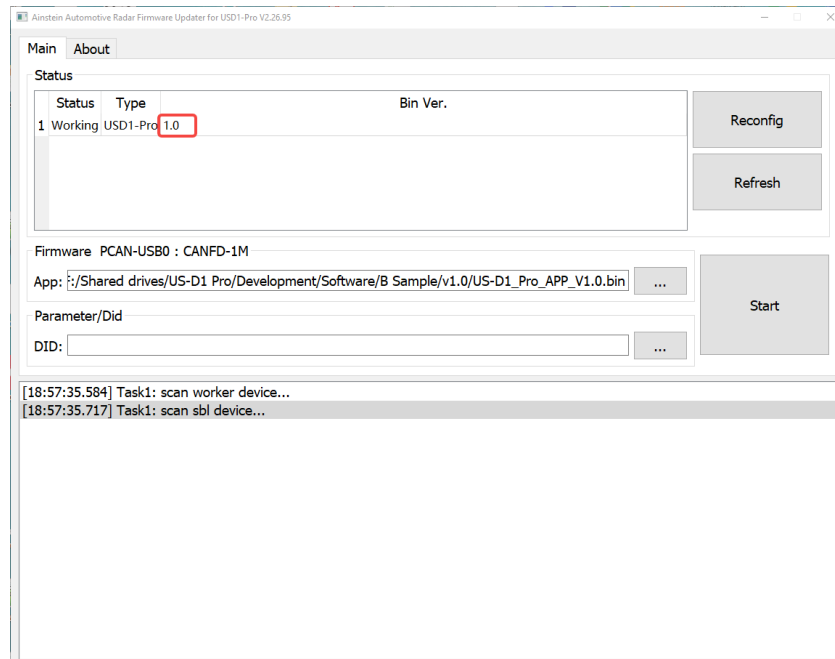
Note:

- Status - Working: US-D1 Pro is operating in the Application
 - Type - USD1-Pro: GUI recognize the device as US-D1 Pro correctly
 - Bin Ver - 0.1.0: Current application firmware version is v0.1.0 (or v0.1)
4. Perform firmware update
 - a. Selecting Application firmware file
 - b. Click 'Start'

- c. Wait until the firmware update process completes. Once it completes and succeeds, it will show the messages.



5. Click 'Refresh' to inquire about US-D1 Pro's new status

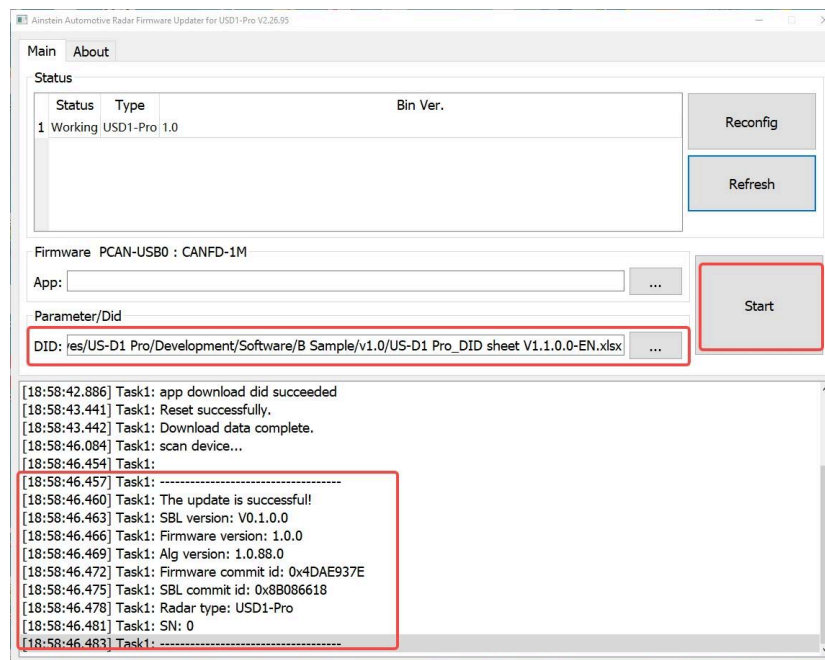


6. Power cycle the US-D1 Pro

8.1.2 Parameter Configuration Instruction

Under private CAN protocol, the configurable parameter information can be found in Section 5.1.2. To perform the parameter configuration, simply use the same GUI tool and follow the steps below.

1. Perform parameter update.
 - a. **Caution:** Clear the content in the App blank, if there is any.
 - b. Select the Data Identifier (DID) sheet file
 - c. Click 'Start'
 - d. Wait until the parameter update process completes. Once it completes and succeeds, it will show the messages.



2. Power cycle the US-D1 Pro

8.1.3 Data Identifier (DID) sheet

In the DID sheet, the mapping between the values in cells and configurable parameters can be found below.

DidTable						
Variable Name	Data Type	Length (Auto)	W/R	Message Typ	Note	Value
crccCode	uint32_t	1	R	E		1
DidTableLength	uint32_t	1	R	E		1
Did_D028	uint8_t	8	RW	E	CAN Configuration	0x1 0x1 0x0 0x0 0x3 0xE8 0x3 0xE8
Did_D029	uint32_t	1	RW	E	Private CAN ID	0x00090002

Note:

1. Only above 5 cells are supported in US-D1 Pro parameter configuration
2. The Cell of 1~5 in above figure is mapping to the No. in the Section 5.1.2 table

3. Number 1,2,4,5 has the drop-down menu, users can select it from there.
4. For CAN 2.0, select the same value for Cell 4 and 5; while for CAN FD, they can be either same value or different.

8.2 DroneCAN Firmware and Parameter Update

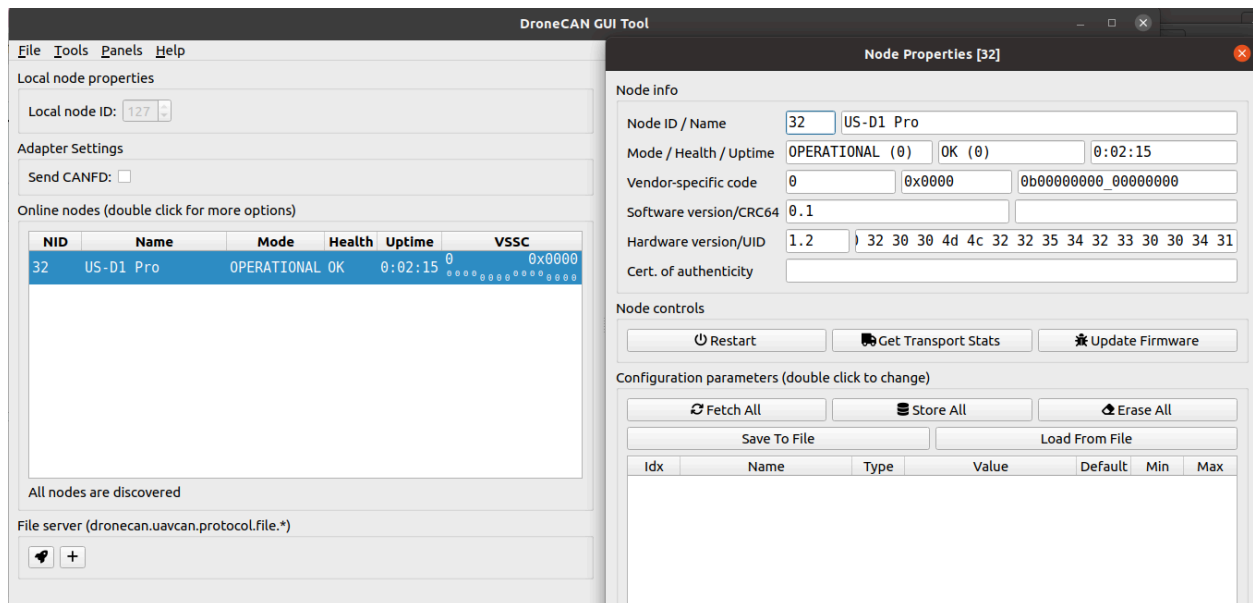
As DroneCAN is the public communication protocol in UAV industry, Ainstein recommend users to use DroneCAN GUI Tool (cross-platform free open source application for DroneCAN bus management and diagnostics) or Mission Planner (versatile ground control station (GCS) software developed by ArduPilot) to perform the firmware and parameter update. Refer to the official instruction for details of steps.

Please see the configurable parameter in Section 5.2.3, and contact Ainstein for more assistance.

8.2.1 DroneCAN GUI tool Firmware Update Instruction

Here is an example of using DroneCAN GUI tool to perform the firmware update.

1. Launch the DroneCAN GUI tool
2. Click US-D1 Pro Node and check all information



3. Perform firmware update
 - a. Click 'Update Firmware'
 - b. Selecting Application firmware file (.bin)

- c. Wait until the firmware update process completes. GUI shows 'FIRMWARE UPDATE'.

The screenshot shows the 'Node Properties [32]' window. The 'Node info' section displays the following fields:

- Node ID / Name: 32 US-D1 Pro
- Mode / Health / Uptime: **FIRMWARE_UPDATE (3)** OK (0) 0:00:11
- Vendor-specific code: 0 0x0000 0b00000000_00000000
- Software version/CRC64: 0.1
- Hardware version/UID: 1.2 32 30 30 4d 4c 32 32 35 34 32 33 30 30 34 31
- Cert. of authenticity: d0 c1 a7 c6 f1 00 00 01 55 40 10 22 fe 00 00 00 18 40

The 'Node controls' section contains three buttons: Restart, Get Transport Stats, and Update Firmware.

The 'Configuration parameters (double click to change)' section includes buttons for Fetch All, Store All, Erase All, Save To File, and Load From File.

At the bottom, a status bar displays the message: 'Firmware update response: uavcan.protocol.file.BeginFirmwareUpdate(error=0, optional_error_me'.

4. Once it completes and succeeds, it will automatically reboot and operate.

Node Properties [32]

Node info

Node ID / Name

32

US-D1 Pro

Mode / Health / Uptime

OPERATIONAL (0)

OK (0)

0:00:13

Vendor-specific code

0

0x0000

0b00000000_00000000

Software version/CRC64

1.0

Hardware version/UID

1.2

32 30 30 4d 4c 32 32 35 34 32 33 30 30 34 31

Cert. of authenticity

Node controls

Restart

Get Transport Stats

Update Firmware

Configuration parameters (double click to change)

Fetch All

Store All

Erase All

Save To File

Load From File

Idx	Name	Type	Value	Default	Min	Max
-----	------	------	-------	---------	-----	-----

Firmware update response: uavcan.protocol.file.BeginFirmwareUpdate(error=0, optional_error_me